Sprint 1

Project Details	
Client	Global Mart Company
Project	Global Mart BI Project
Start Data	
End Date	
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1. Sprint 1 Overview

Sprint Name	Sprint 1 – Foundation Architecture & Platform Baseline	
Sprint Timeline	Sprint Duration	2 weeks
	Sprint Start Date	
	Sprint End Date	
Sprint Goal	Design and implement the minimum viable data platform foundation that enables reliable data ingestion into raw storage layer, supported by clear architectural and tooling decisions.	
In Scope	This sprint focuses on architecture definition, foundational tooling decisions, repository structure, and ingestion of a single dataset into the raw layer as proof of concept	
Out of Scope	• Dimensional modeling • Business transformations • Analytics dashboard • Workflow orchestration or scheduling • Production hardening	
Sprint Output	• ✅ Documented architecture • ✅ Infrastructure baseline: MinIO, MySQL, and Hive Metastore on Kubernetes, with a smoke test • 🔄 Raw data ingestion: in progress, carried over to the next sprint	

2. Inputs & Assumptions

2.1 Inputs from Sprint 0	
Input	**Description**
Business Events	Approved BEAM outputs
Data Sources	Defined and validated
Data Requirements	Approved high-level data quality expectations (accuracy, completeness, consistency, timeliness, validaty)
Stakeholders	Identified and agreed

2.2 Assumptions & Constraints	
Assumption	Batch ingestion only
Assumption	One dataset in Sprint 1
Constraint	2-week sprint
Constraint	Open-source tooling

3. Target Architecture

3. 1 Architecture Layers		
Layer	**Purpose**	**Status**
Source	Operational & external systems	Existing
Ingestion	Move data into platform	In progress (basic/manual)
Raw (Bronze)	Store data as-is	Storage ready, data not yet loaded
Clean (Silver)	Standardize & clean data	Deferred
Curated (Gold)	Analytics-ready data	Deferred

3.2 Architecture Principles	
Principle	**Description**
Separation of Concerns	Each layer has a single responsibility
Schema-on-Read	Flexibility for evolving schemas
Incremental Evolution	Platform grows sprint by sprint
Reproducibility	All steps documented

4. Technology & Tooling Decisions

4.1 Analytical Storage (Data Lake)	
Aspect	**Decision**
Selected Technology	MinIO (S3-compatible object storage)
Purpose	Store raw and analytical datasets
Data Type	Parquet files
Reason	Scalable, decoupled, analytics-optimized
Alternatives Considered	Relational DB, HDFS
Rejected Because	Tight coupling(DB) and operational complexity (HDFS)
Deferred	Advanced table formats (Iceberg/Delta)

4.2 Metadata & Operational Storage	
Aspect	**Decision**
Selected Format	MySQL
Purpose	Metadata, schemas, catalog
Used By	Hive Metastore
Reason	Mature, reliable, transactional
Not Used For	Analytical fact storage

5. Repository Structure

5.1 Storage Decision	
Path	**Purpose**
/docs	Sprint documentation
/data/raw	Raw ingested datasets
/infra	Infrastructure config
/scripts	Ingestion scripts
The repository structure reflects the physical implementation, while data layers represent	

logical architecture.

6. Sprint 1 Implementation Summary

6.1 Dataset Selection	
Aspect	**Decision**
Dataset Name	Brazilian E-Commerce (hosted in SQL Server)
Source	Kaggle / External
Reason	High business relevance

6.2 Ingestion Details	
Aspect	**Decision**
Ingestion Type	Batch
Method	Manual / Script
Target Layer	Raw (Bronze)
Validation	Basic data sanity checks (schema, row count, sample rows)

7. Definition of Done

Item	status	why
Architecture documented	✅ Done	Section 3 of the doc, plus DESIGN.md
Tooling decisions justified	✅ Done	Section 4 gives reasons and rejected alternatives
Repo structure finalized	✅ Done, **after one fix**	Section 5 still lists /infra, /data/raw, /scripts, but your repo now has docker/, docs/, infrastructure/, datasets/. Update section 5 to the real layout, then mark it done.
Raw ingestion successful	🔄 In progress	The export is drafted but not validated, and nothing has landed in MinIO
Validation checks passed	🔄 In progress	In this doc it means data checks (schema, row count, sample rows), which can't happen

		before ingestion works
Platform smoke test passed	✅ Done	Smoke-test Job passes: DNS and TCP checks for MySQL and MinIO (v0.1.0 release notes)
Risks documented	✅ Done	Section 8

8. Risks, Trade-Offs & Deferred Items

Item	Description
Risk	Manual ingestion not scalable
Trade-off	Speed over automation
Deferred	Silver/Gold layers
Deferred	Orchestration